

EE4C10

Analog Circuit Design Fundamentals (References)

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Electronic Instrumentation Laboratory (EI)

LTspice setup

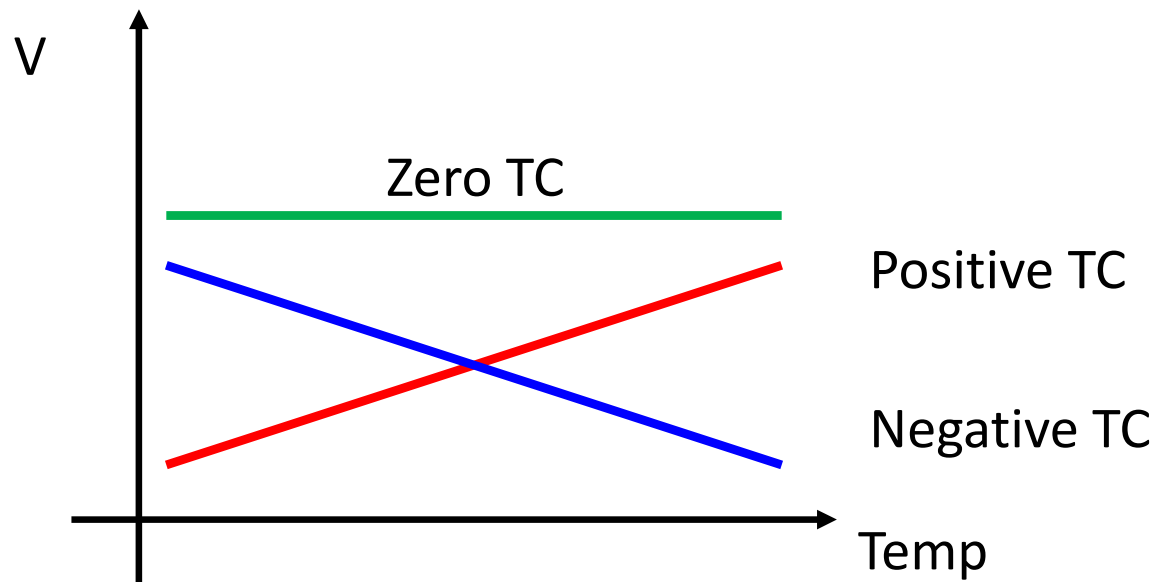
- Copy the file 'log018_mod.l' to your LTspice folder.
- Open the file, search for 'log018_mod.l' and modify the path accordingly.
- It is probably going to be like

```
.lib 'C:\Program Files\LTC\LTspiceXVII\lib\cmp\log018_mod.l' BIP
```

- Fix all 6 paths.

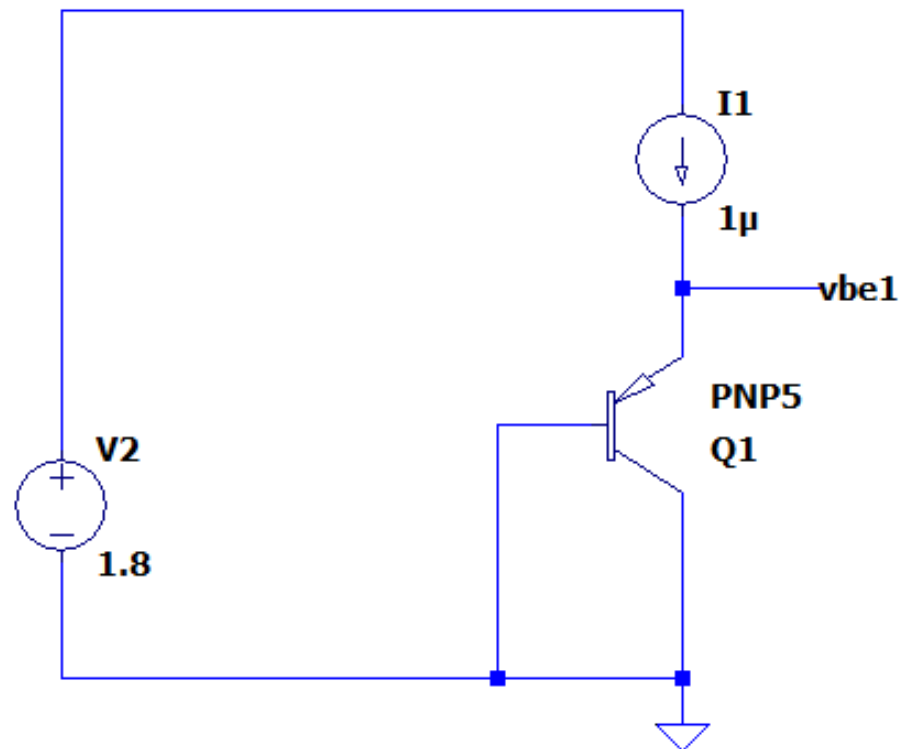
Bandgap reference principle

- Generate voltages with positive and negative temperature coefficients (TCs).
- Add them with proper weighting -> zero TC.



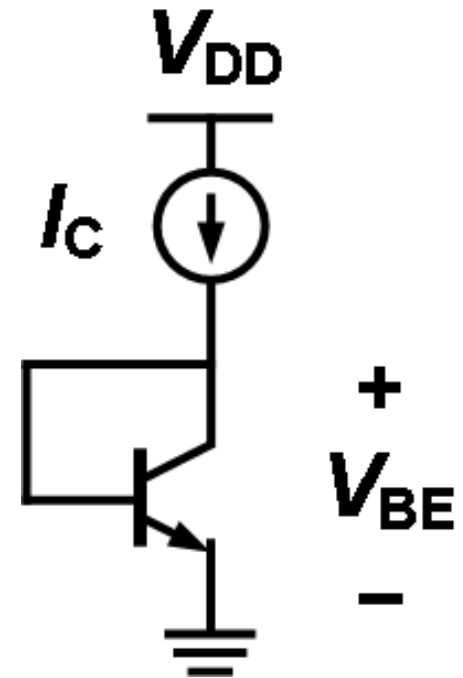
E1. BJT: Negative-TC voltage generation

- How does V_{be} vary over temperature and corners?
- Does this match the estimation in Razavi's book?



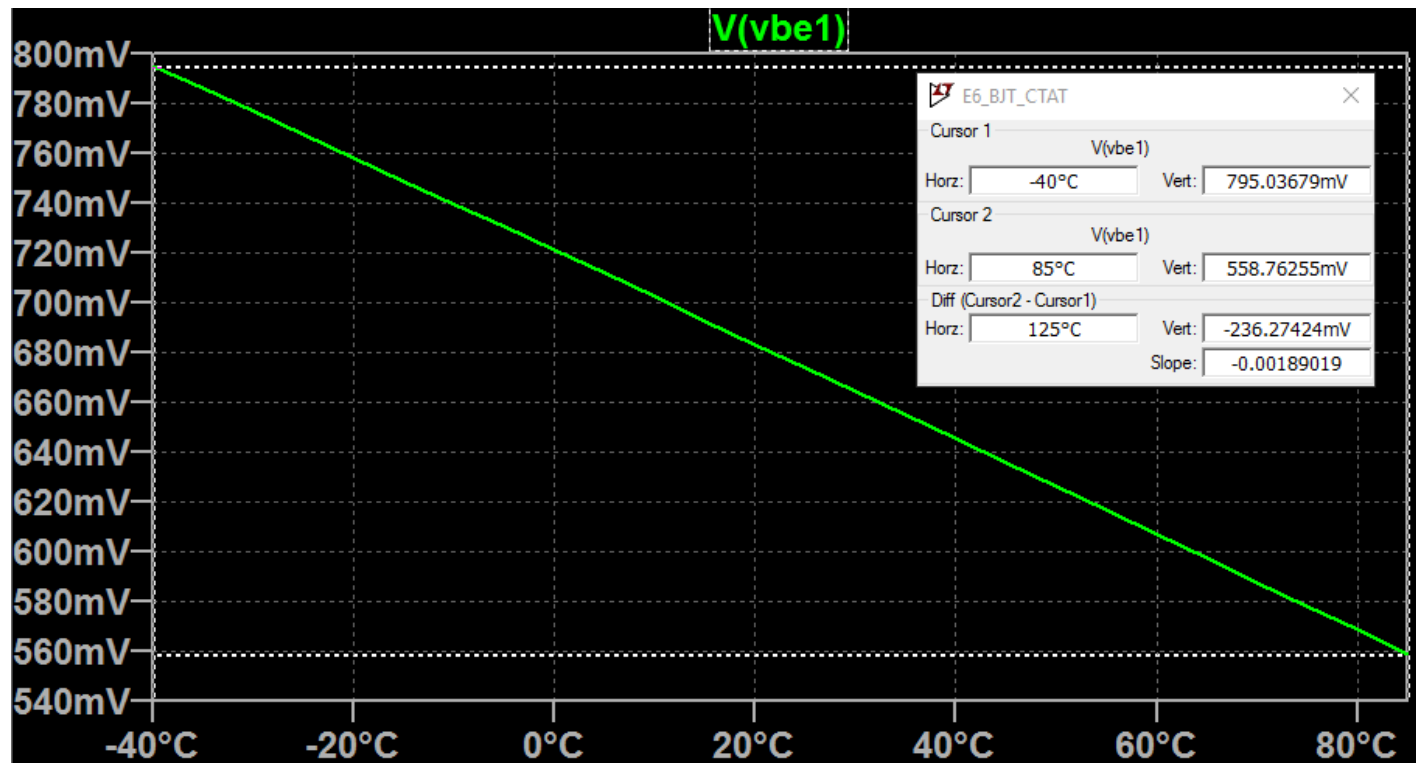
E1. BJT: Negative-TC voltage generation

- $V_{BE} = V_T \cdot \ln\left(\frac{I_C}{I_S}\right)$
- Assuming a constant I_C ,
- $\frac{\partial V_{BE}}{\partial T} \approx -1.5\text{mV}/^\circ\text{C}$ with $V_{BE} \approx 750\text{mV}$
(CTAT voltage)
- A lot of approximation involved!



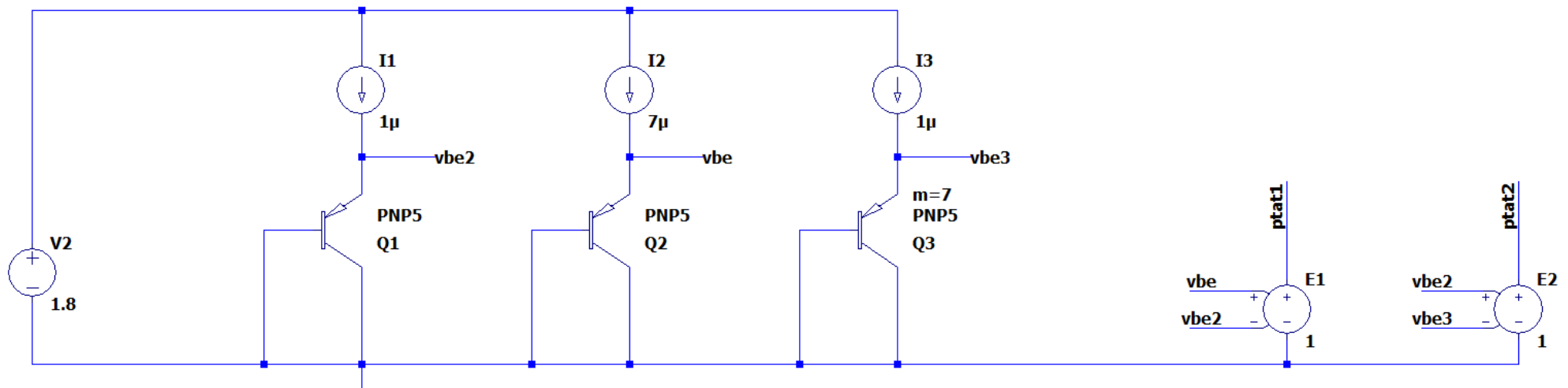
E1. BJT: Negative-TC voltage generation

- Estimation in Razavi's book: $\frac{\partial V_{BE}}{\partial T} \approx -1.5\text{mV}/^\circ\text{C}$
- Simulation result: $-1.89\text{mV}/^\circ\text{C}$



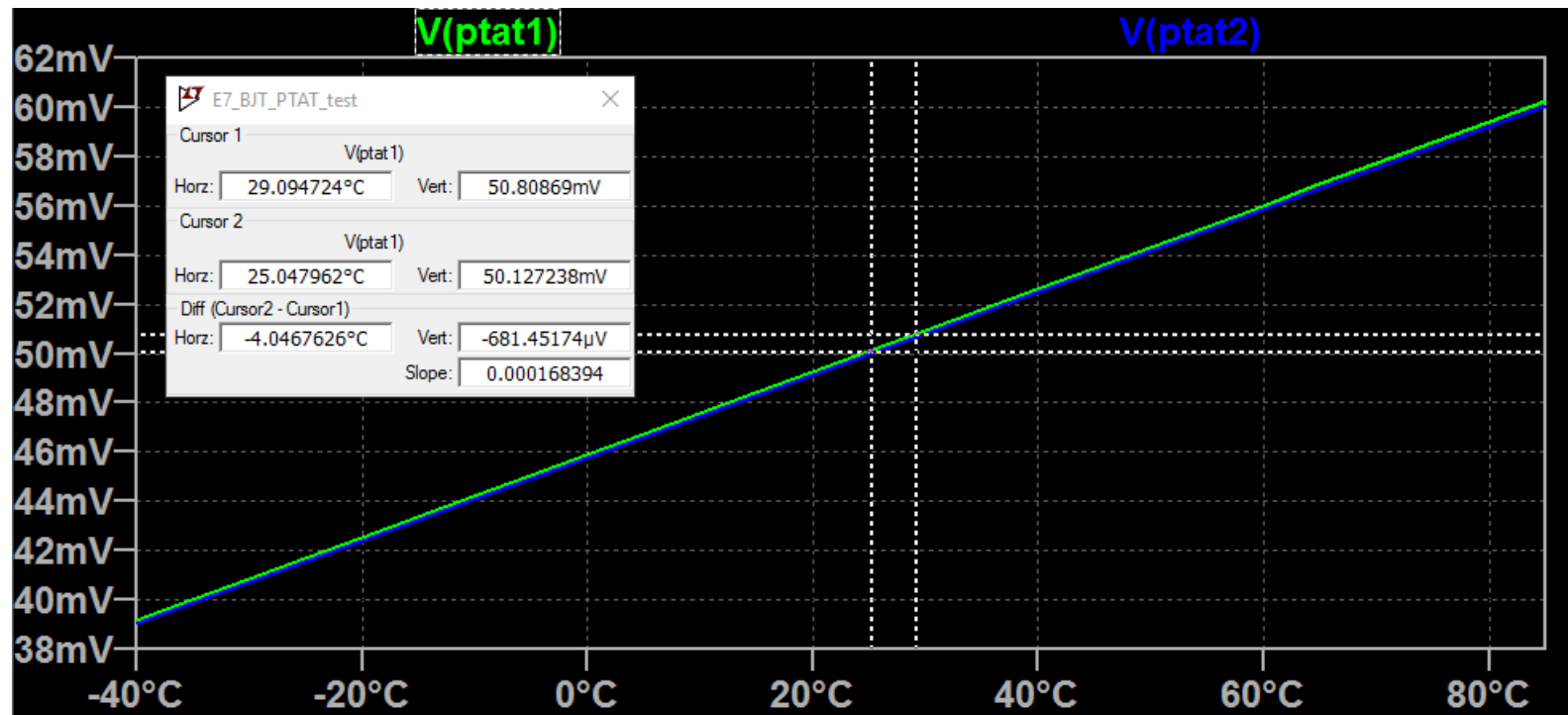
E2. BJT: Positive-TC voltage generation

- Two ways of generating a PTAT voltage
- Does the result match the one in Razavi's book?
- Is there any significant difference between the two?



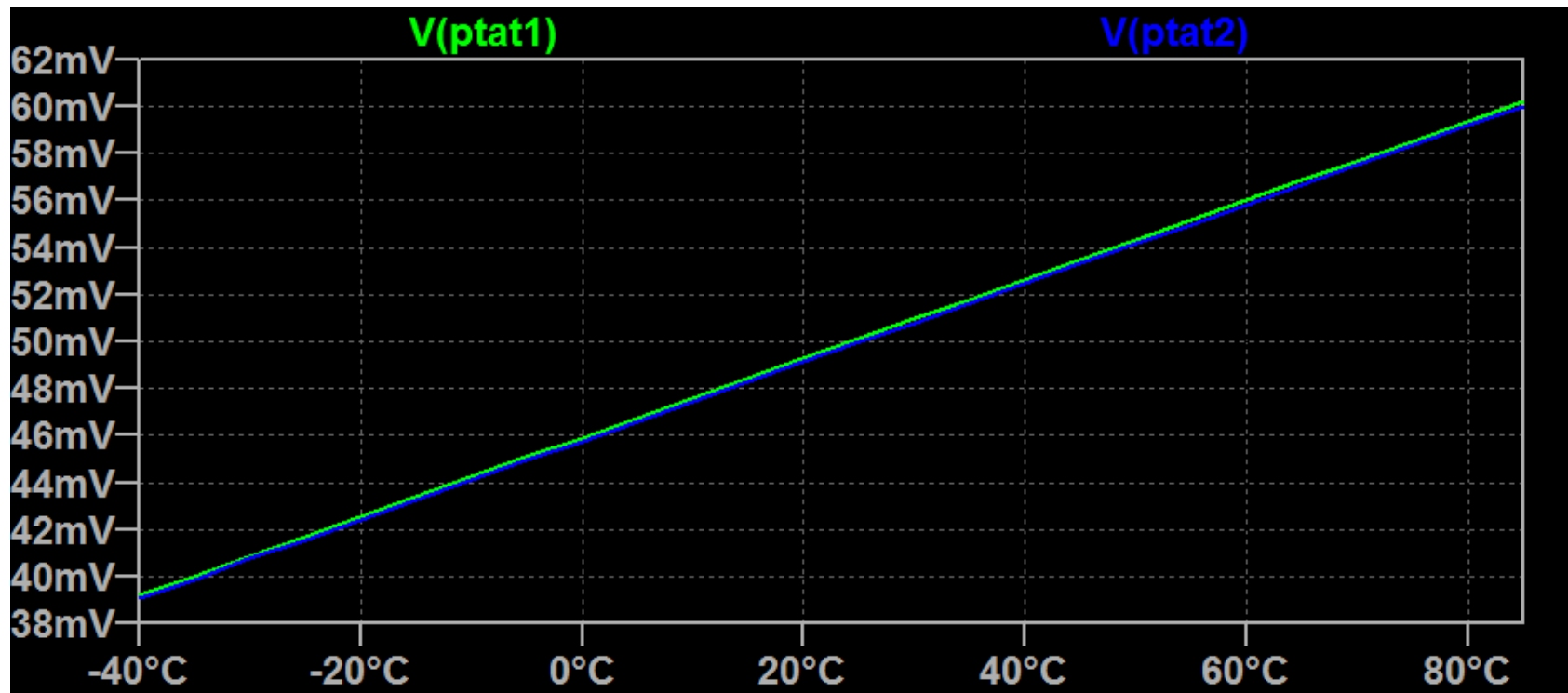
E2. BJT: Positive-TC voltage generation

- Model: $\Delta V_{PTAT}/\Delta T = k/q \cdot \ln n = 168 \mu\text{V}/^\circ\text{C}$
- Simulation: also $168 \mu\text{V}/^\circ\text{C}$



E2. BJT: Positive-TC voltage generation

- Similar results despite a 7x difference in current density. This PTAT voltage is pretty accurate & stable!
- You can see how V_{CTAT} changes with a 7x larger current.

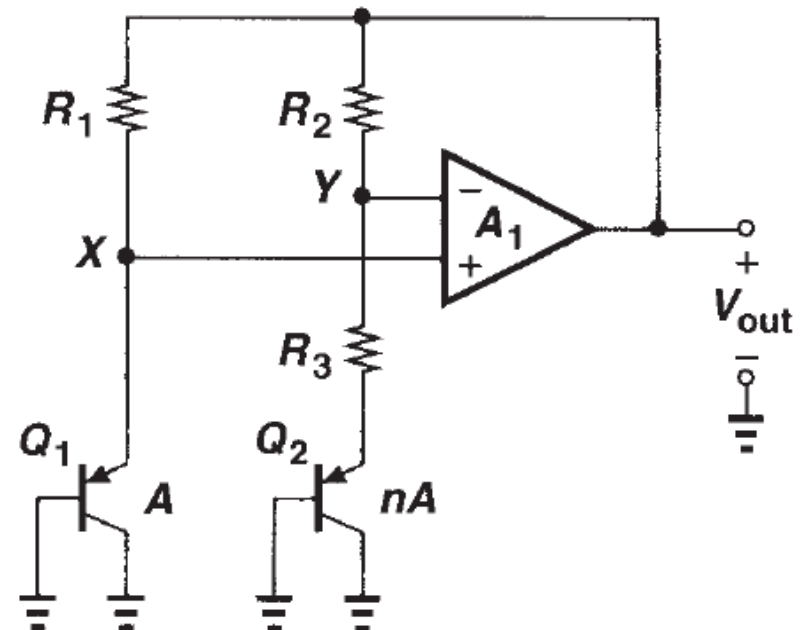


E3. Bandgap voltage reference

- $R_1 = R_2$
- Opamp forces $V_X = V_Y$
- $V_{out} = V_{BE2} + V_T \ln n \cdot \left(1 + \frac{R_2}{R_3}\right)$

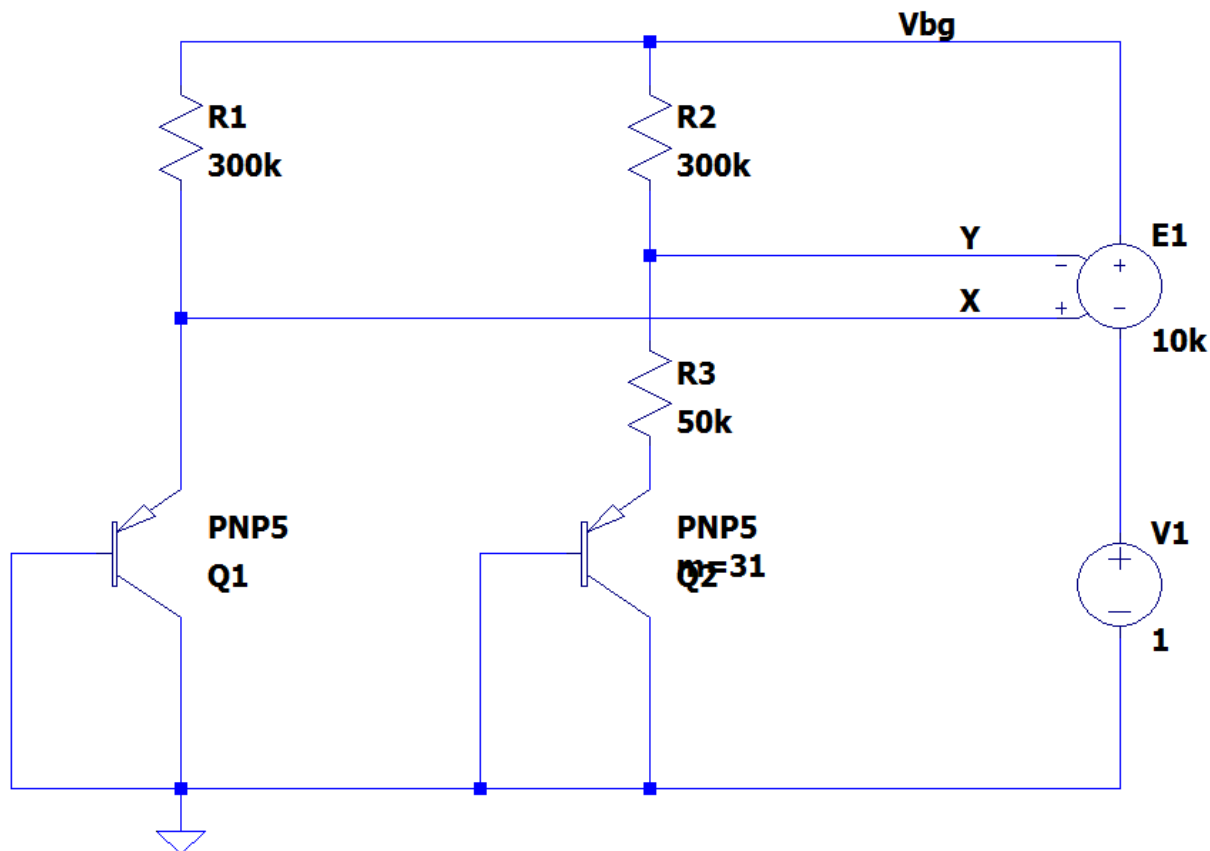
V_{CTAT}

V_{PTAT}



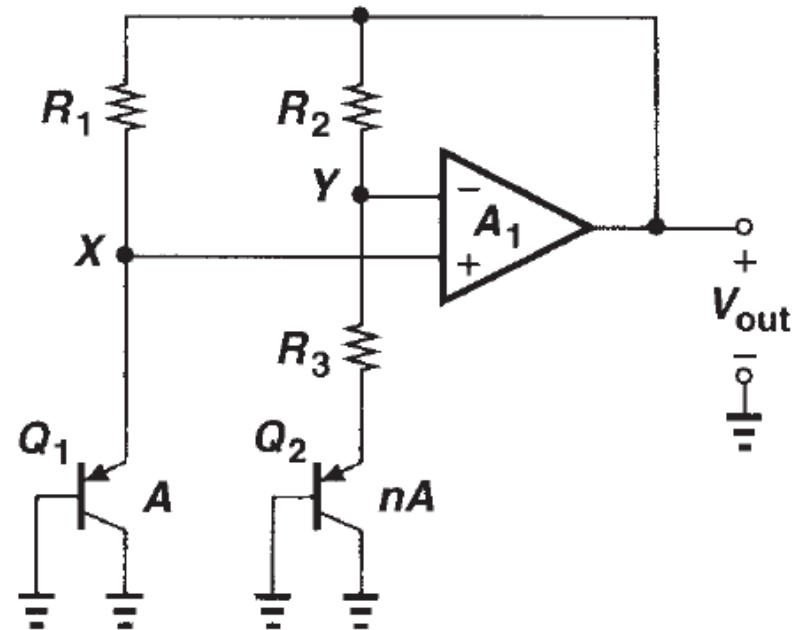
E3. Bandgap voltage reference

- Simulate this bandgap.
- How to scale V_{PTAT} and V_{CTAT} ?



E3. Bandgap voltage reference

- $V_{out} = V_{BE2} + V_T \ln n \cdot \left(1 + \frac{R_2}{R_3}\right)$
- Estimation in Razavi's book
 $(1 + R_2/R_3) \ln n \approx 17.2$.
- Simulation result
 $(1 + R_2/R_3) \ln n \approx 24.0$
- Why? Think about the previous simulation results?



E3. Bandgap voltage reference

- Now simulate the circuit under different corners.

```
.lib 'C:\Program Files\LTC\LTspiceXVII\lib\cmp\log018_mod.l' TT  
.lib 'C:\Program Files\LTC\LTspiceXVII\lib\cmp\log018_mod.l' TT_BIP
```



```
.lib 'C:\Program Files\LTC\LTspiceXVII\lib\cmp\log018_mod.l' SS  
.lib 'C:\Program Files\LTC\LTspiceXVII\lib\cmp\log018_mod.l' SS_BIP
```

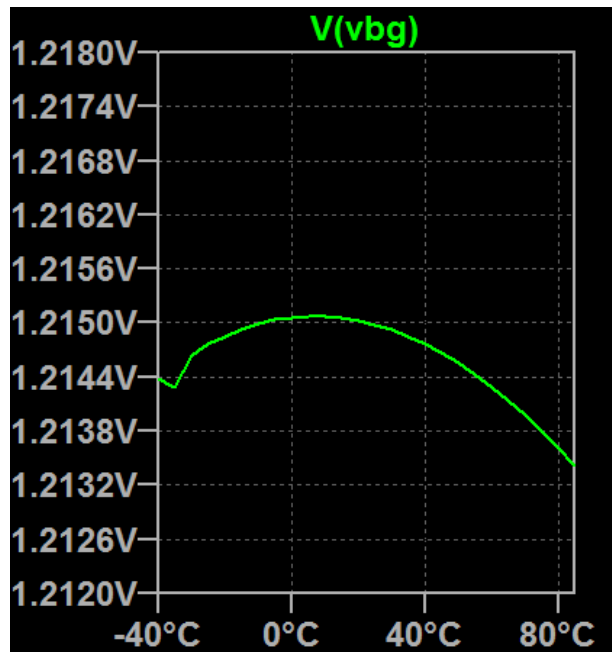
OR

```
.lib 'C:\Program Files\LTC\LTspiceXVII\lib\cmp\log018_mod.l' FF  
.lib 'C:\Program Files\LTC\LTspiceXVII\lib\cmp\log018_mod.l' FF_BIP
```

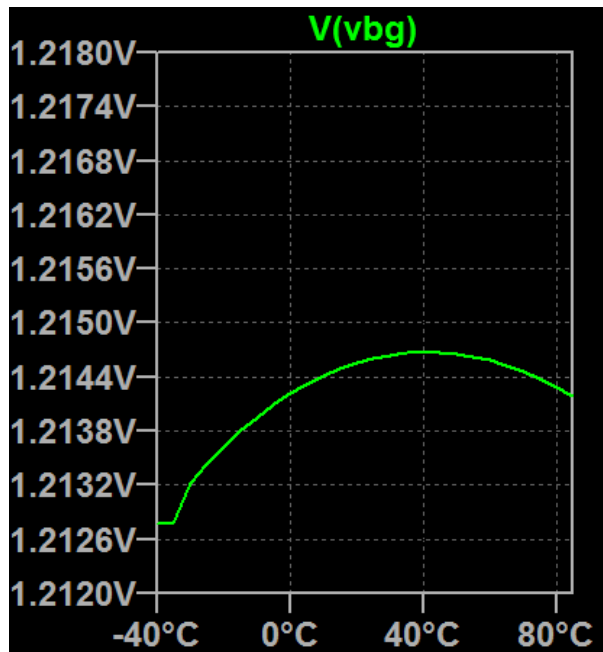
E3. Bandgap voltage reference

- Variation over corners
- Also a challenge in your homework!

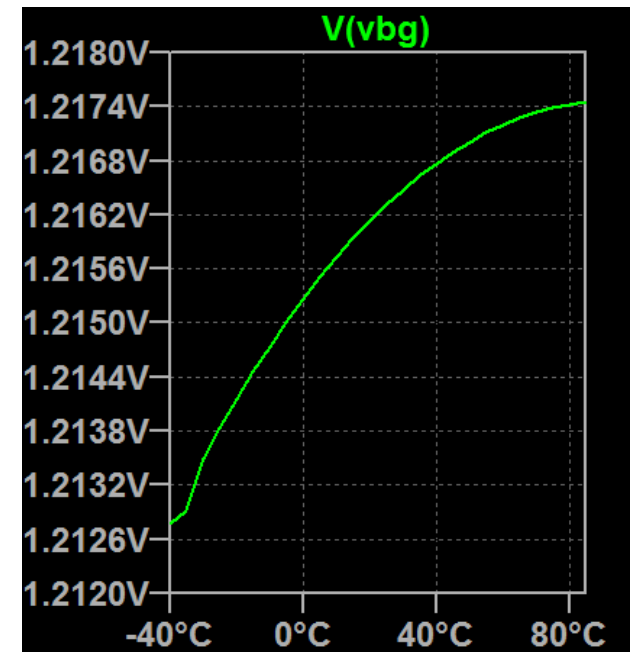
FF



TT

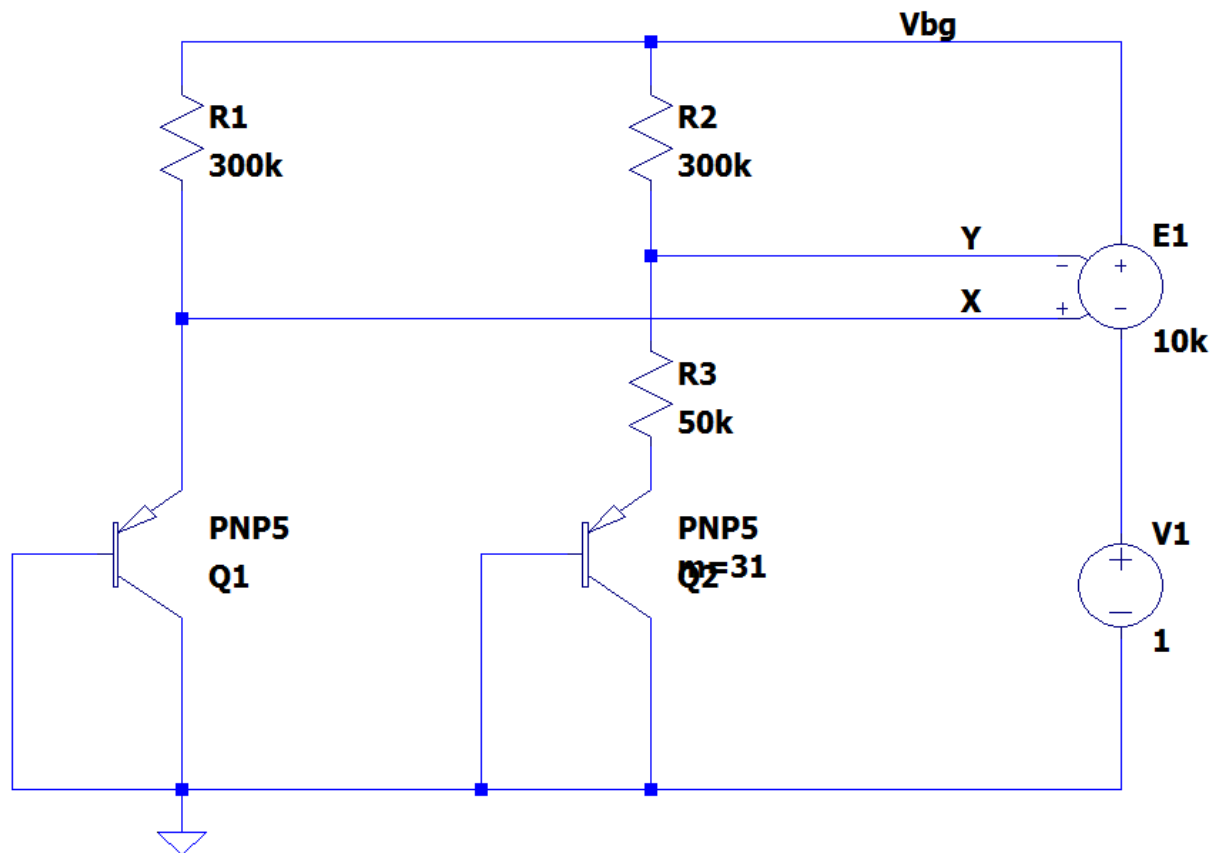


SS



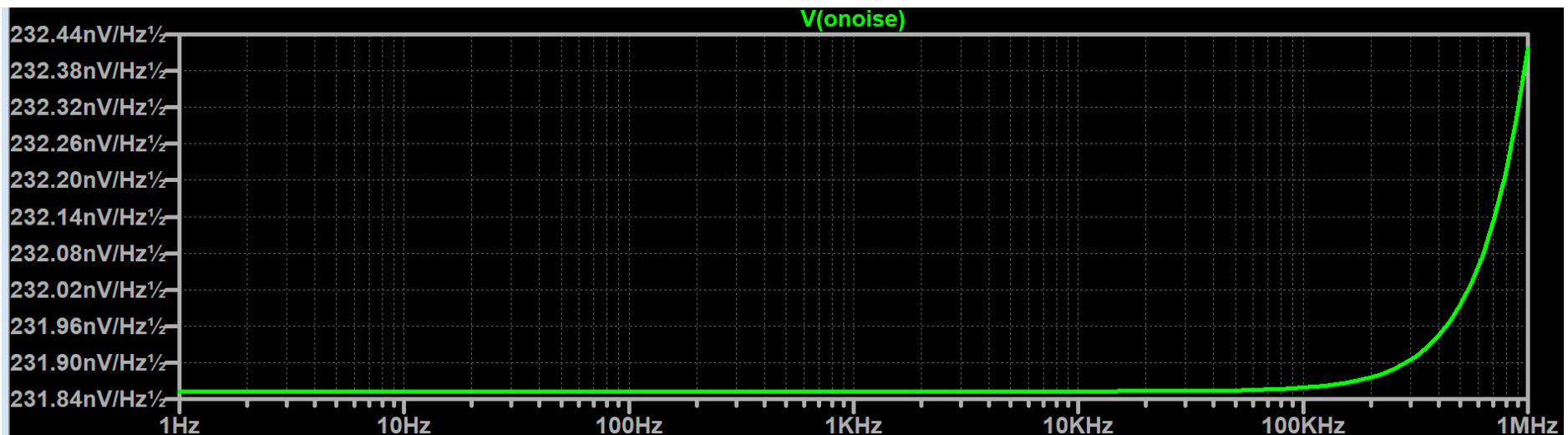
E4. Bandgap voltage reference: Noise

- Simulate the output noise
- Compare it with that of a 300k Ω resistor



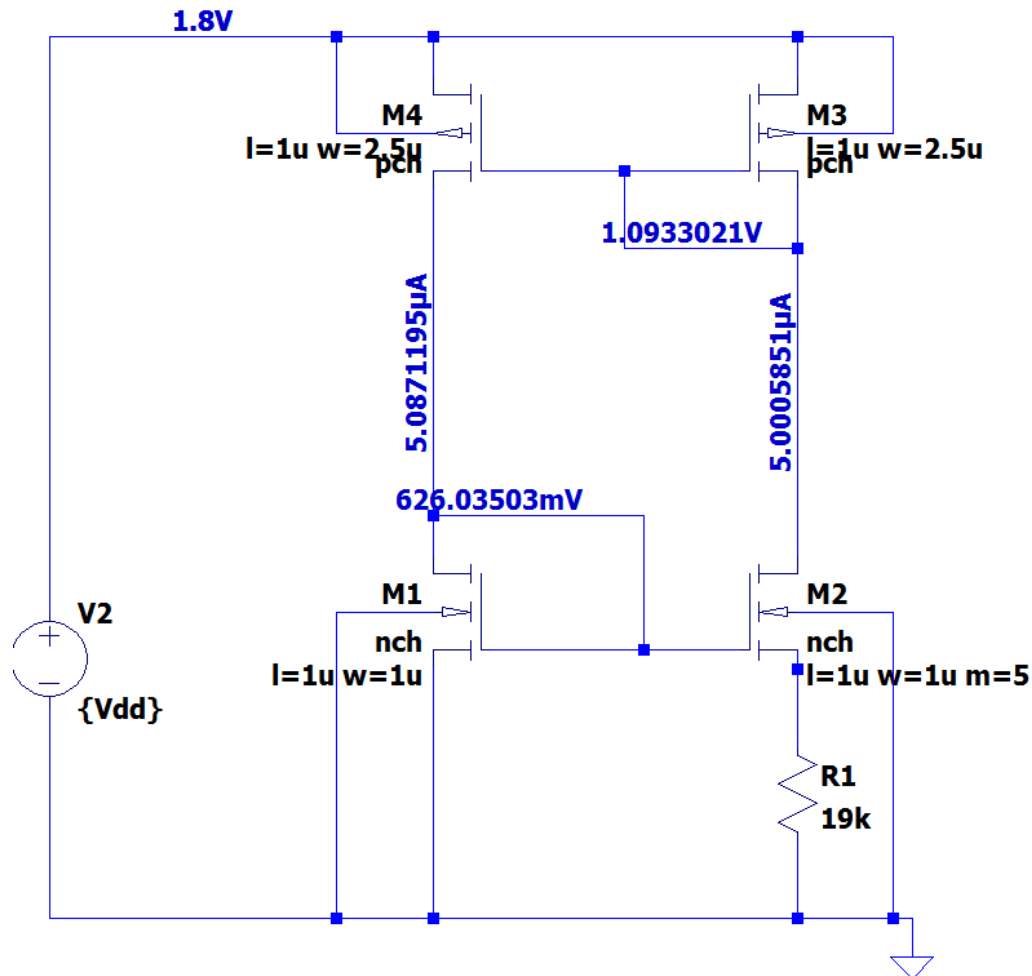
E4. Bandgap voltage reference: Noise

- Simulated output noise density: $\sim 232\text{nV}/\sqrt{\text{Hz}}$
- Noise density of a $300\text{k}\Omega$ resistor: $\sqrt{4kTR} \approx 70\text{nV}/\sqrt{\text{Hz}}$
- What is the dominant noise source? why?



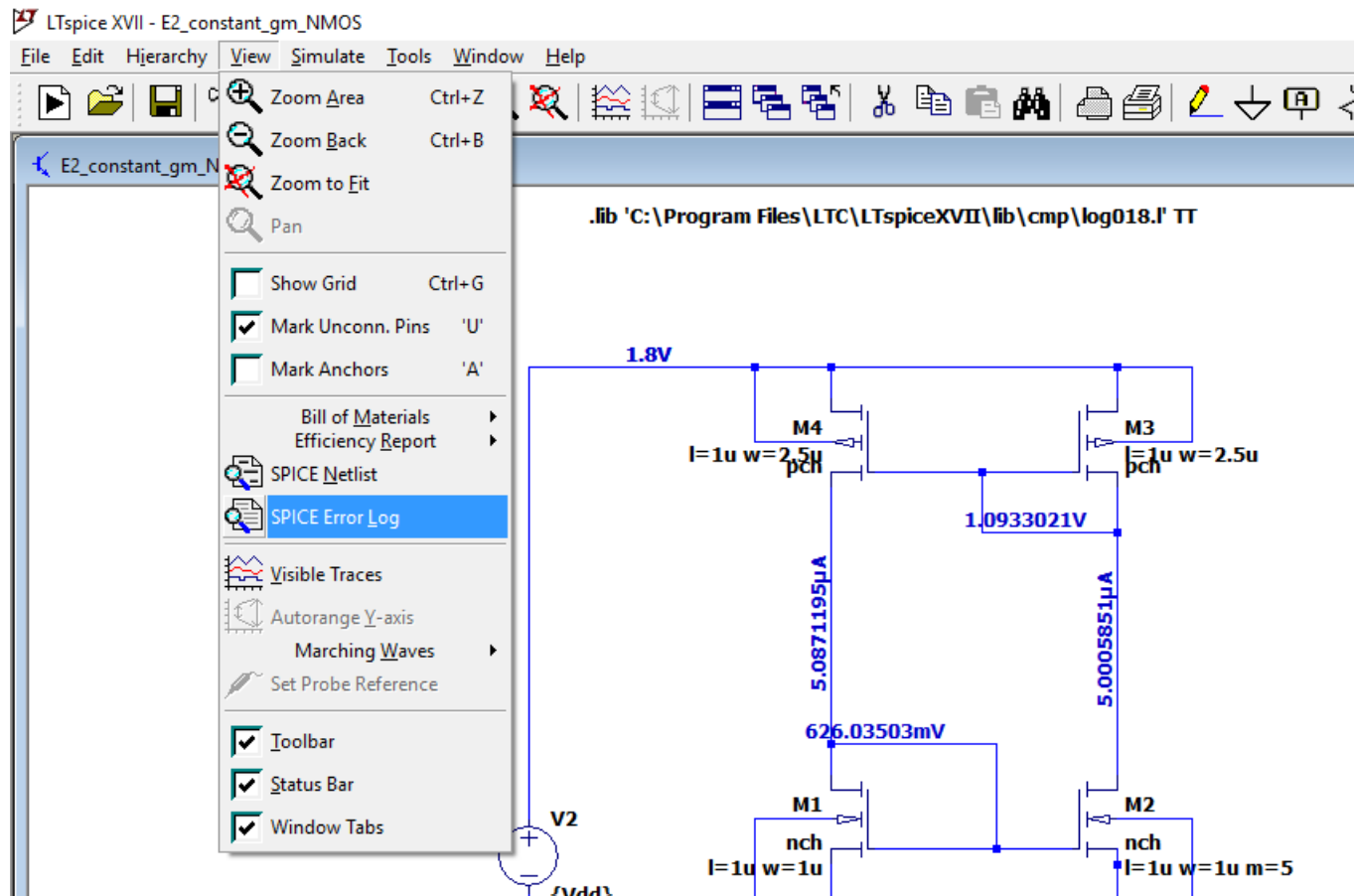
E5. Constant-gm biasing, NMOS

- What is the expected gm of M1?
- What is the gm of M1 over PVT? (1.6V to 2V; -40°C to 85°C)



E5. Constant-gm biasing, NMOS

- What is the expected gm of M1?
- What is the gm of M1 over PVT? (1.6V to 2V; -40°C to 85°C)



E5. Constant-gm biasing, NMOS

- Theoretical $g_{m1} = \frac{2}{R_S} \left(1 - \frac{1}{\sqrt{K}} \right) \approx 58.2 \mu\text{S}$
- Actual gm value: slightly smaller

gm1	-40°C	27°C	85°C
Vdd=1.6V	54.4μS	51.9μS	48.8μS
Vdd=1.8V	55.3μS	52.5μS	49.3μS
Vdd=2.0V	56.2μS	53.2μS	49.7μS

11%

2.5%